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CAPSTONE PROJECT

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Student / Group Template

Project Title	Campus Help Desk Ticketing System: A Database-Driven Application for IT Support Request Management
Group Members	Freek Schaefer-Gerber and Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Requirements finalised
Submission Date	29 October 2026

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1. Purpose and Scope

Define the product and its boundaries.

The system centralizes customer issues into trackable tickets for customers, agents and administrators. In scope: account access, ticket submission, tracking, assignment, comments, status changes and basic reporting. Out of scope for the MVP: payment processing, AI classification, live chat, SLA automation and third-party integrations.

2. Users and Stakeholders

Describe user classes and stakeholder needs.

Customer: submits and monitors own tickets. Agent: handles assigned work. Administrator: manages assignments and users. Project team: implements and tests. Lecturer: reviews capstone outcomes.

3. Functional Requirements

Use IDs FR-01, FR-02 etc. Include description, priority and acceptance criterion.

ID	Requirement	Priority	Acceptance criterion
FR-01	Users can register and sign in securely	Must	Valid user can log in; invalid credentials rejected
FR-02	Customer creates ticket with subject, category, description and priority	Must	Valid ticket saved with unique ID
FR-03	Customer views own tickets and status history	Must	Other customers' tickets are not visible
FR-04	Agent views and updates assigned tickets	Must	Changes persist with actor and timestamp
FR-05	Administrator assigns tickets to agents	Must	Assigned agent can access ticket
FR-06	Customer and agent exchange ticket comments	Must	Comments shown chronologically to authorized users
FR-07	Admin manages user roles and	Should	Authorized admin can modify

	ticket categories		categories and roles
FR-08	Dashboard displays ticket counts by status and priority	Should	Counts match stored tickets
FR-09	Users search/filter tickets by status and category	Should	Matching records shown
FR-10	System records creation and status-change timestamps	Must	Audit fields populated automatically
FR-11	Customer may attach an image to a ticket	Could	Accepted safe file is linked to ticket
FR-12	Agent can resolve ticket and customer can confirm closure	Should	Ticket transitions recorded

4. Non-Functional Requirements

Use IDs NFR-01 etc. Include measurable criteria.

ID	Attribute	Measurable or testable criterion
NFR-01	Security	Passwords hashed; authorization enforced server-side for each ticket operation
NFR-02	Privacy	Customers can only access their own tickets; no real customer data in demonstrations
NFR-03	Usability	Core ticket submission can be completed on mobile and desktop without horizontal scrolling
NFR-04	Performance	Under local test conditions, 95% of typical ticket-list requests complete within 2 seconds with 100 sample tickets
NFR-05	Reliability	Invalid inputs yield

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		controlled errors without corrupting existing tickets
NFR-06	Maintainability	Separate UI, API/business logic and persistence layers; version control all code
NFR-07	Data integrity	Primary/foreign keys, required fields and permitted status values validated
NFR-08	Accessibility	Forms have visible labels, keyboard navigation and readable contrast

5. Use Cases / User Stories

Document actor, preconditions, main flow, alternatives and postconditions.

UC-01 Create Ticket

Actor: Customer. Precondition: authenticated customer. Main flow: open new ticket form, enter subject/category/description/priority, submit; system validates and saves a new Open ticket with a unique ID. Alternative: missing required field produces an inline error. Postcondition: ticket visible in customer list.

UC-02 Assign Ticket

Actor: Administrator. Precondition: unassigned or reassignable ticket exists. Main flow: select ticket and agent, save; system records assignment. Alternative: invalid agent is rejected. Postcondition: agent sees ticket in assigned queue.

UC-03 Resolve Ticket

Actor: Agent. Precondition: agent authorized for ticket. Main flow: add comment, change status to Resolved, save; system records change. Alternative: unauthorized access denied. Postcondition: customer sees updated status.

6. Data Requirements

Describe key data entities, validation and retention.

The implemented relational database contains three core tables: User (id, name, email, password_hash, role, created_at); Ticket (id, customer_id, assigned_to, title, description, category, priority, status, created_at, updated_at); and Comment (id, ticket_id, user_id, body, created_at). Foreign keys enforce referential integrity. All test data is synthetic.

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keys and uniqueness of user email. Use fictitious data in test environments. Retention and deletion rules require stakeholder confirmation.

7. Interfaces

Describe UI, API, database and external interfaces.

Frontend: responsive browser interface. API: authenticated HTTPS endpoints for users, tickets, assignments, comments and reports. Database: relational SQL access through backend only. External services: none required for the MVP.

8. Constraints and Assumptions

Document technical, legal, operational and project constraints.

Two-student capstone; submission deadline 29 October 2026. Suggested stack is not yet verified. Assume one customer owns each ticket and one agent may be assigned at a time. Email notifications and file uploads are optional. Performance targets apply to a controlled local test, not production guarantees.

9. Acceptance Criteria

Define what must be true for the solution to be accepted.

- All Must requirements pass documented test cases
- Customers cannot access another customer’s tickets
- Agent and admin permissions are demonstrated
- Tickets can be created, viewed, assigned and updated
- Source code, database schema and screenshots are supplied
- Critical defects are resolved or explicitly documented

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